TRANSISTOR

THEORY OF OPERATION

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

LABORATORY

50%

EXAM

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Transistor Construction and Operation

Transistor Configurations

- Common Base
- Common Collector
- Common Emitter

Characteristic Curve

Regions of Operation

Reading Datasheet

TRANSISTOR CONSTRUCTION AND OPERATION

THE FIRST TRANSISTOR

December 23, 1947

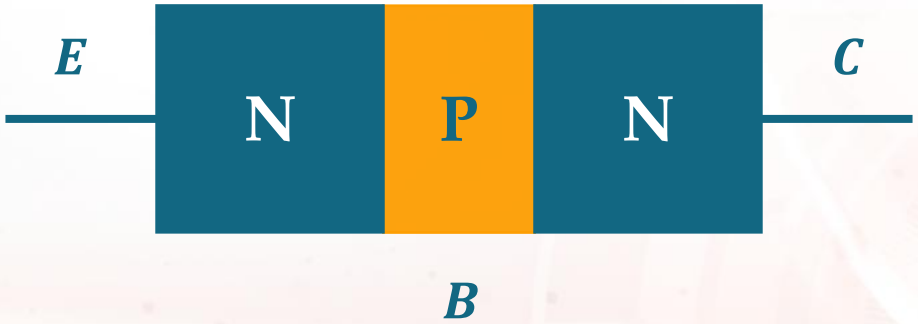
Dr. William Shockley

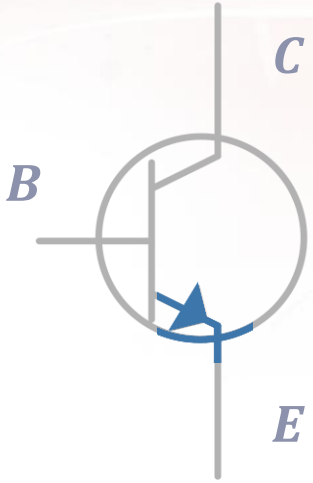
Dr. John Bardeen

Dr. Walter Brattain

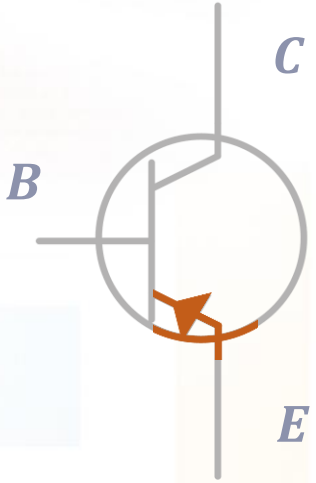
CONSTRUCTION

NPN

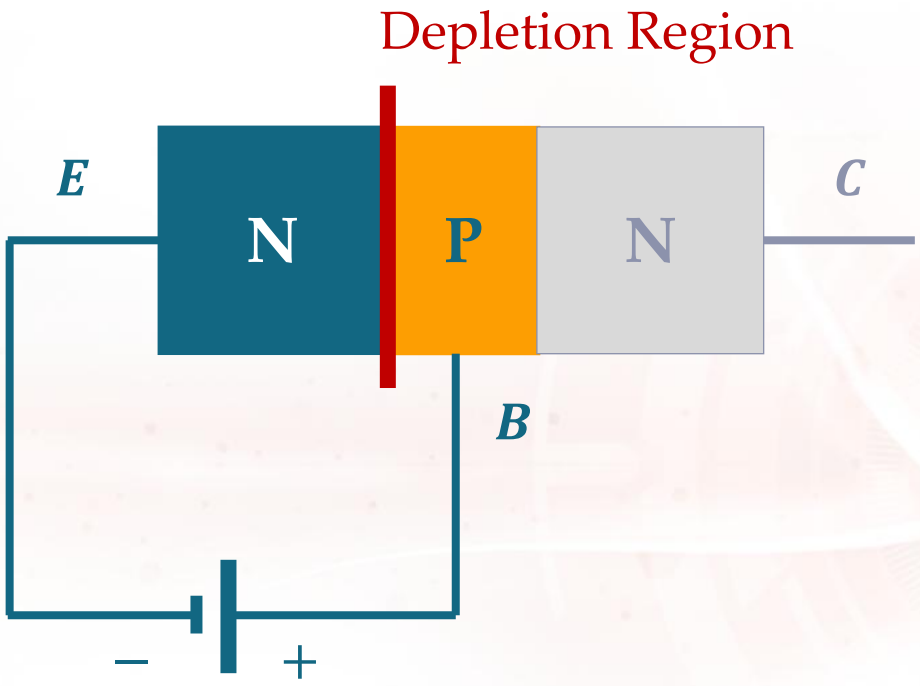




PNP



EMITTER-BASE DIODE

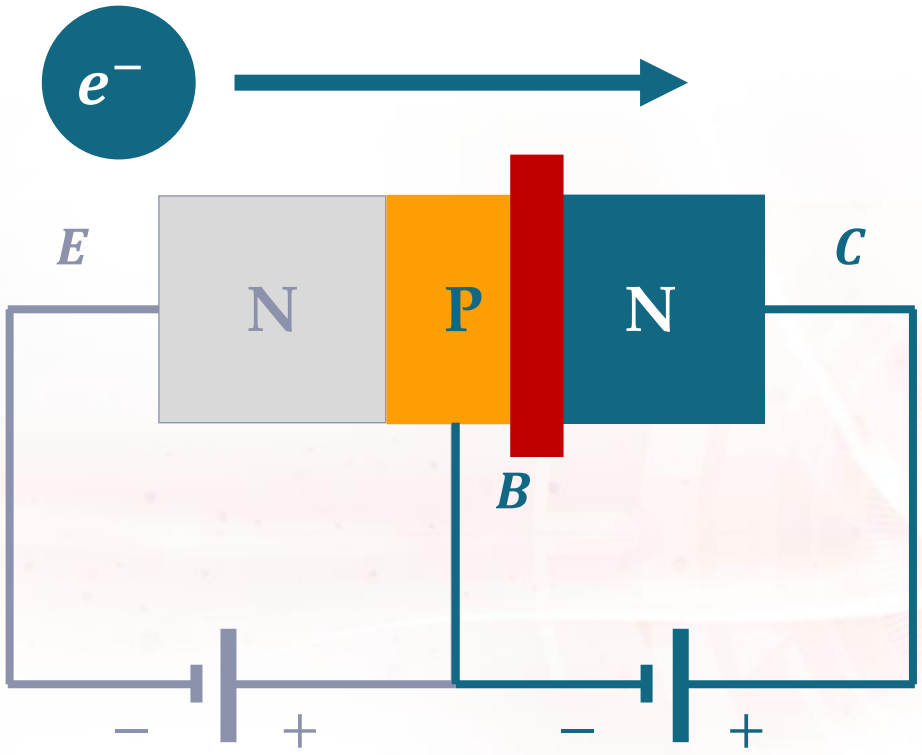


forward-biased

Emitter is **heavily** doped

Base is **lightly** doped

COLLECTOR-BASE DIODE

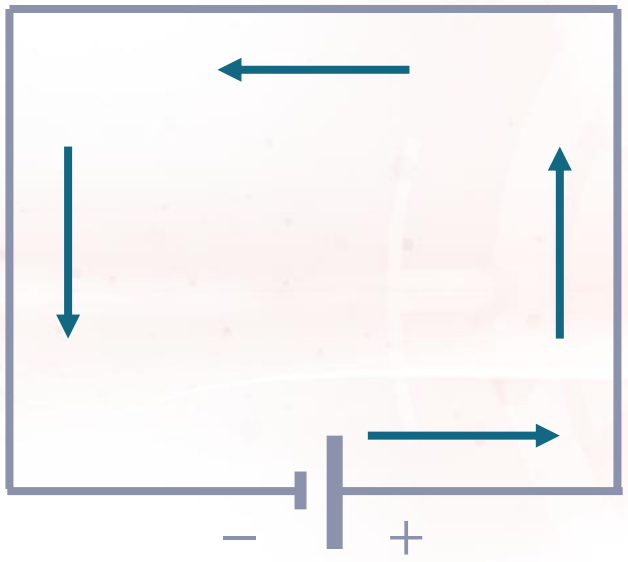


reverse-biased

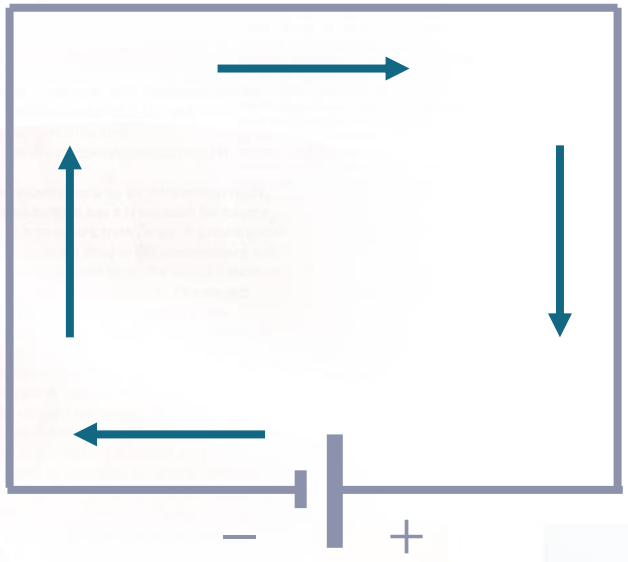
Collector is doped at an **intermediate** level

CURRENT FLOW NOTATION

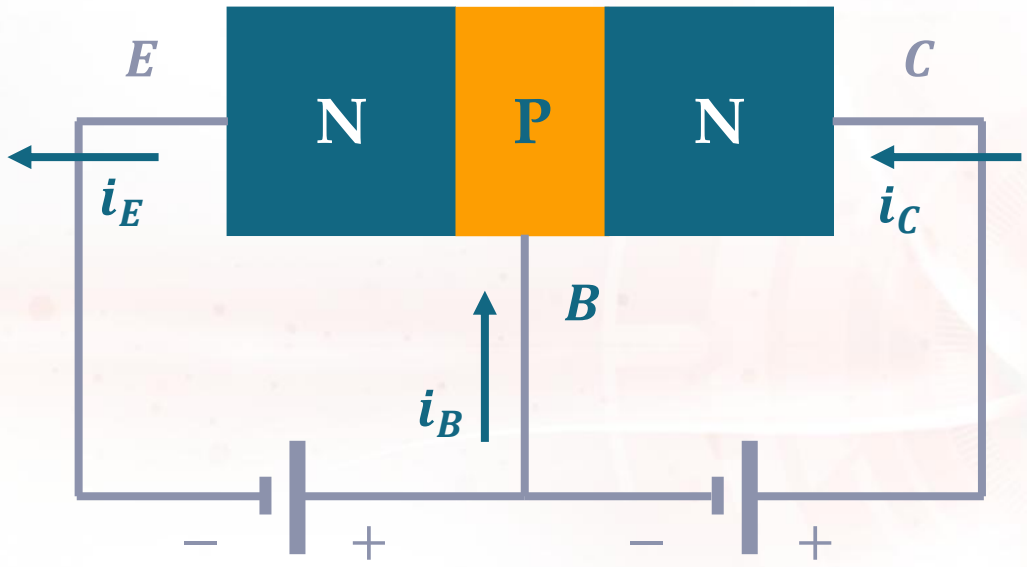
Conventional Flow Notation



Electron Flow Notation



KIRCHHOFF'S CURRENT LAW



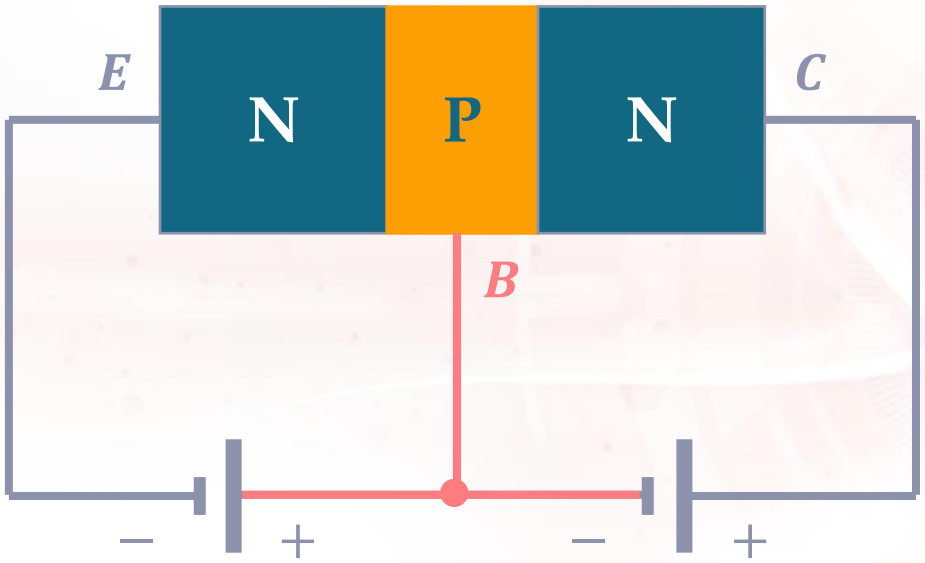
KCL @ B

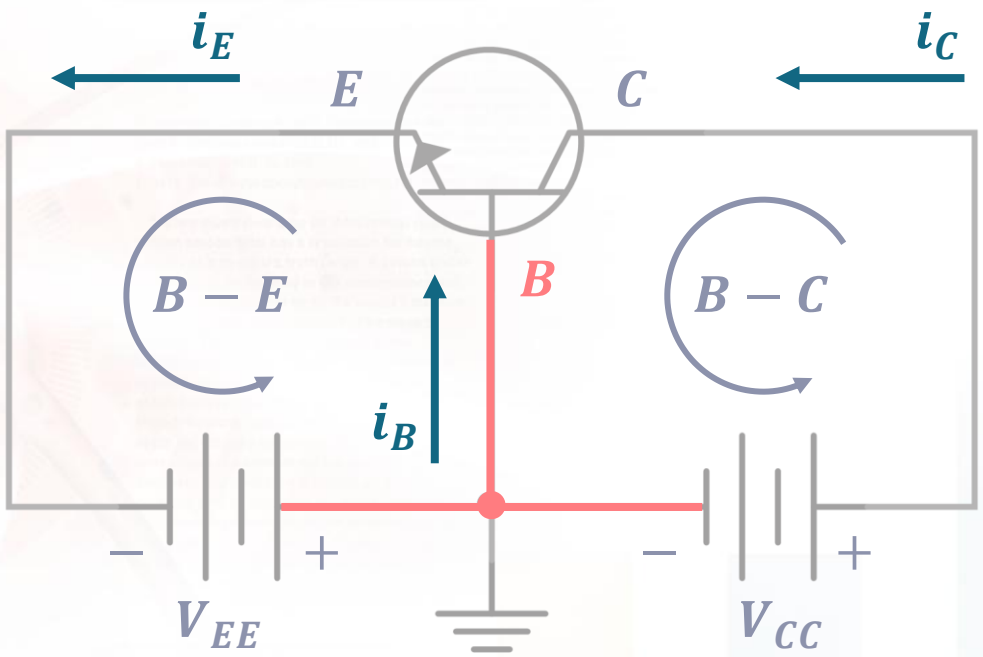
$$-i_C - i_B + i_E = 0$$

$$i_E = i_C + i_B$$

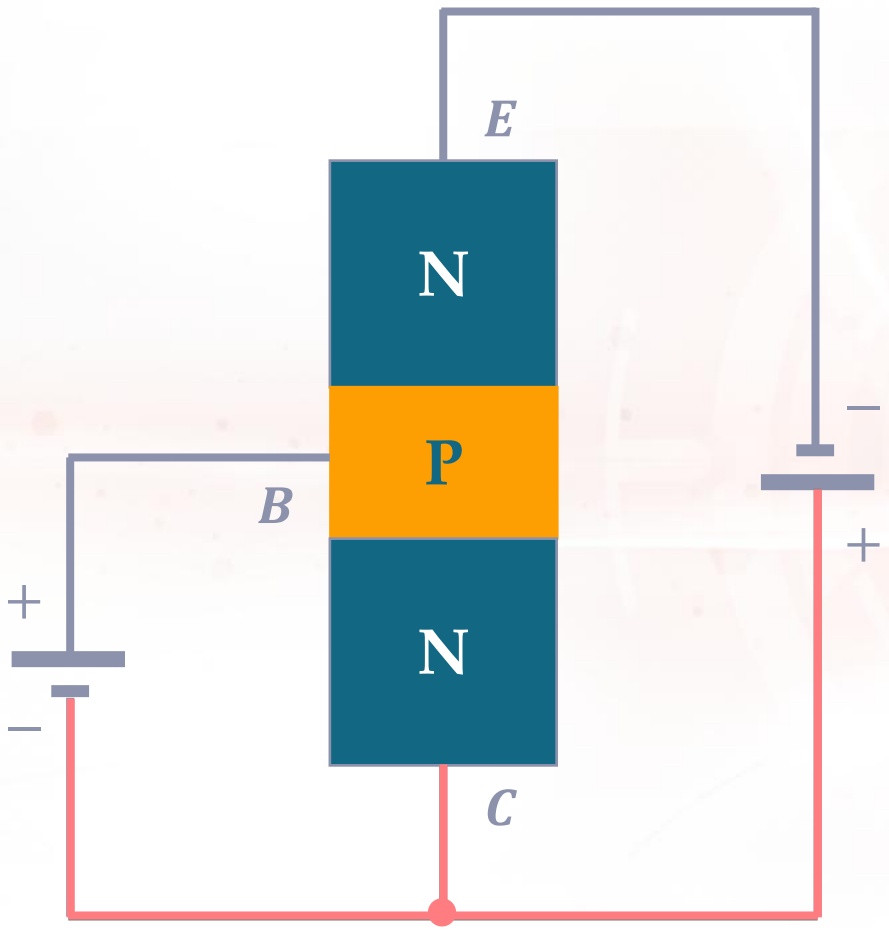
TRANSISTOR CONFIGURATIONS

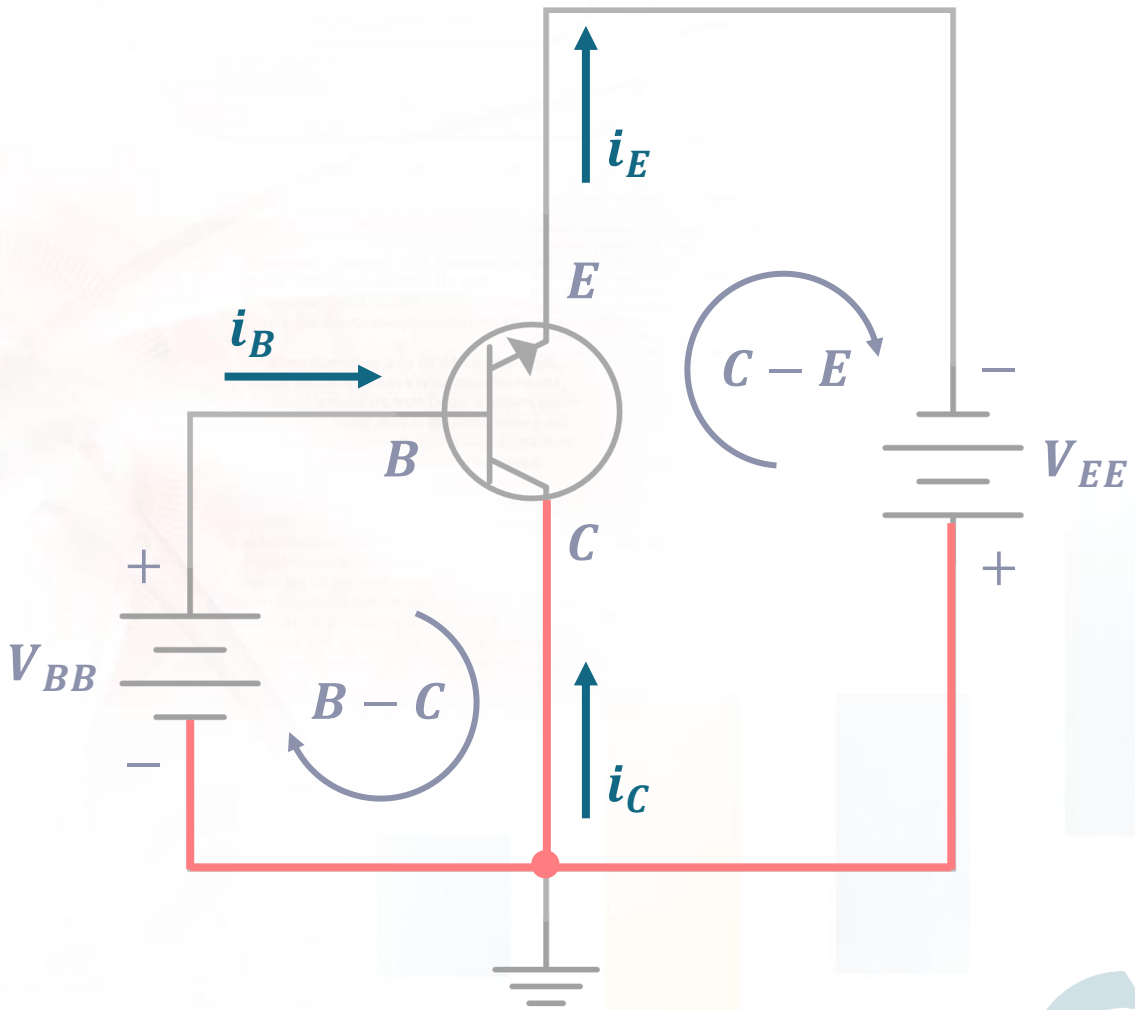
COMMON-BASE



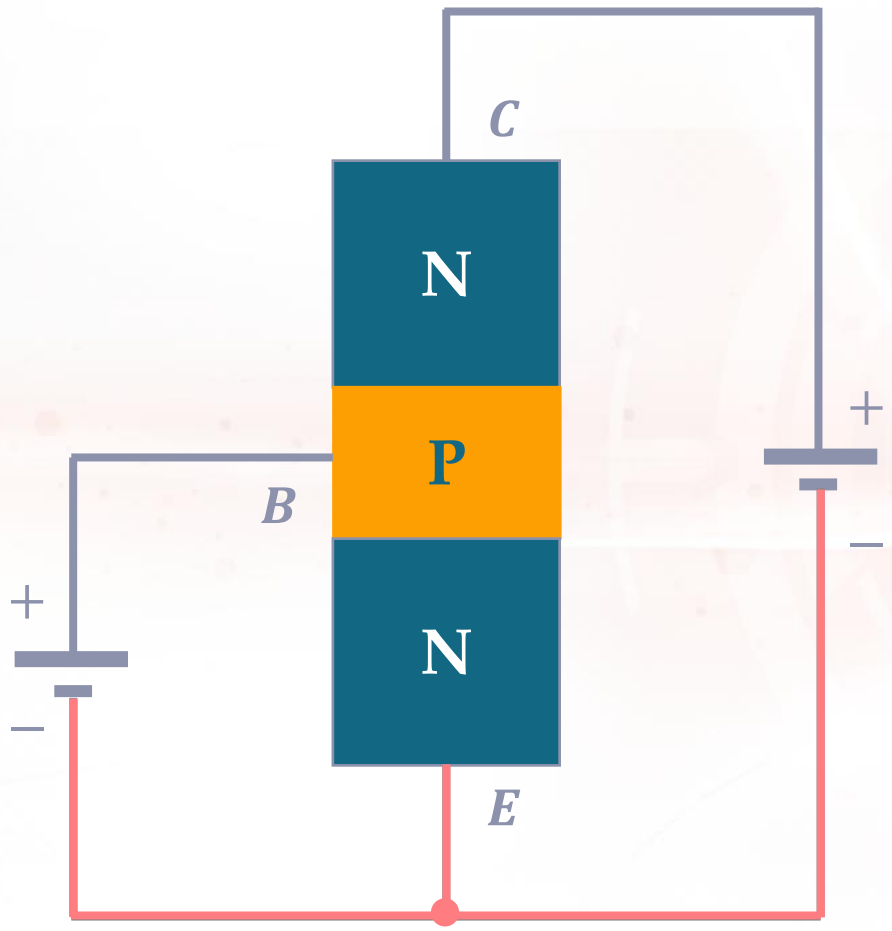


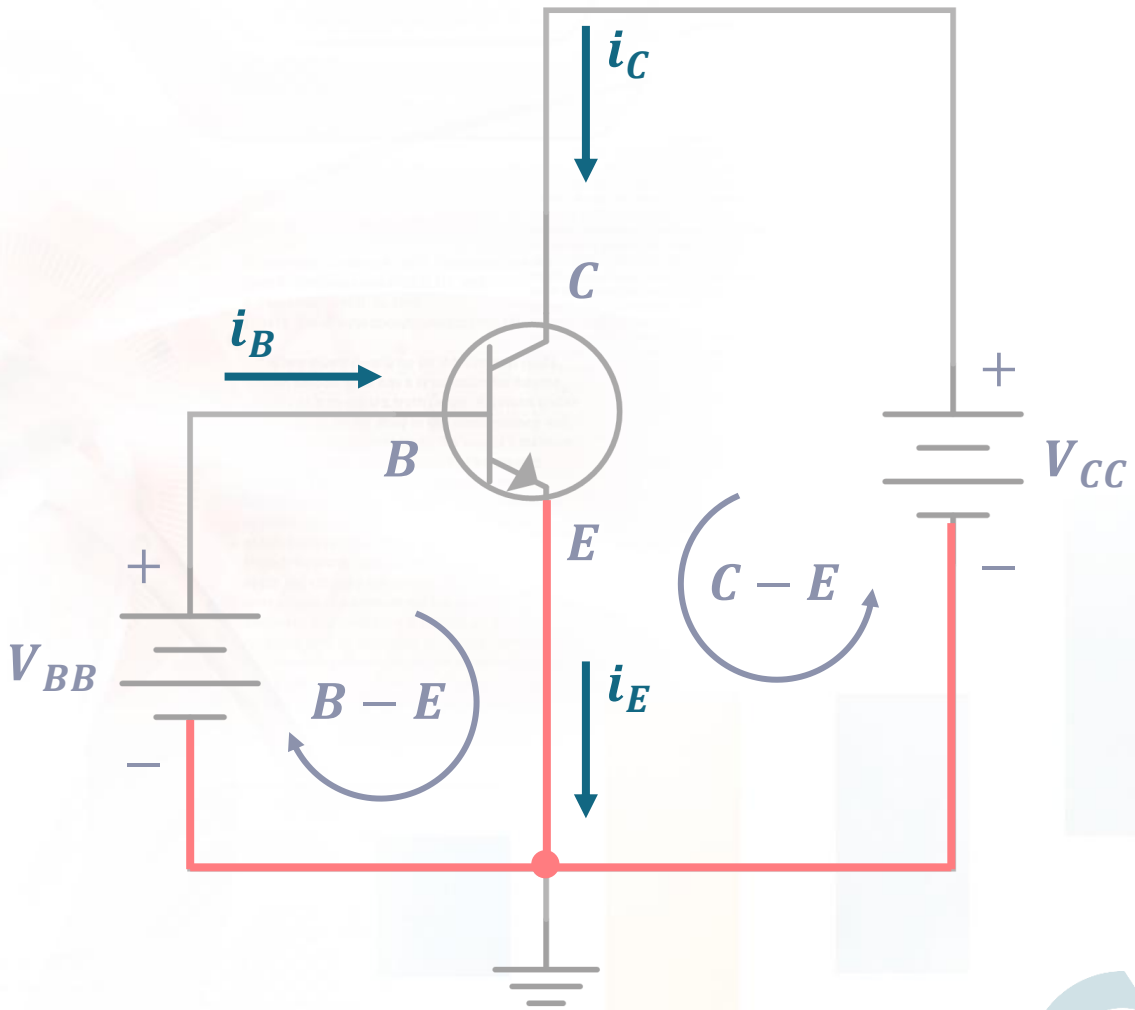
COMMON-COLLECTOR





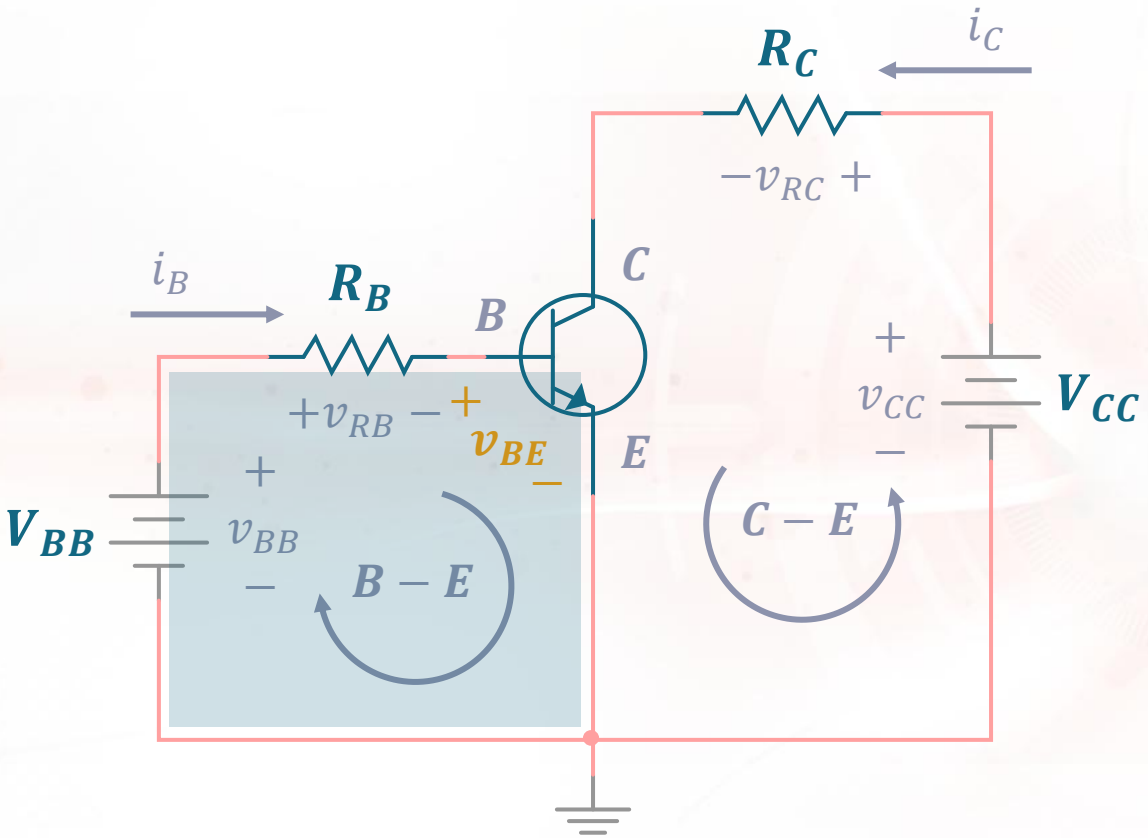
COMMON-EMITTER

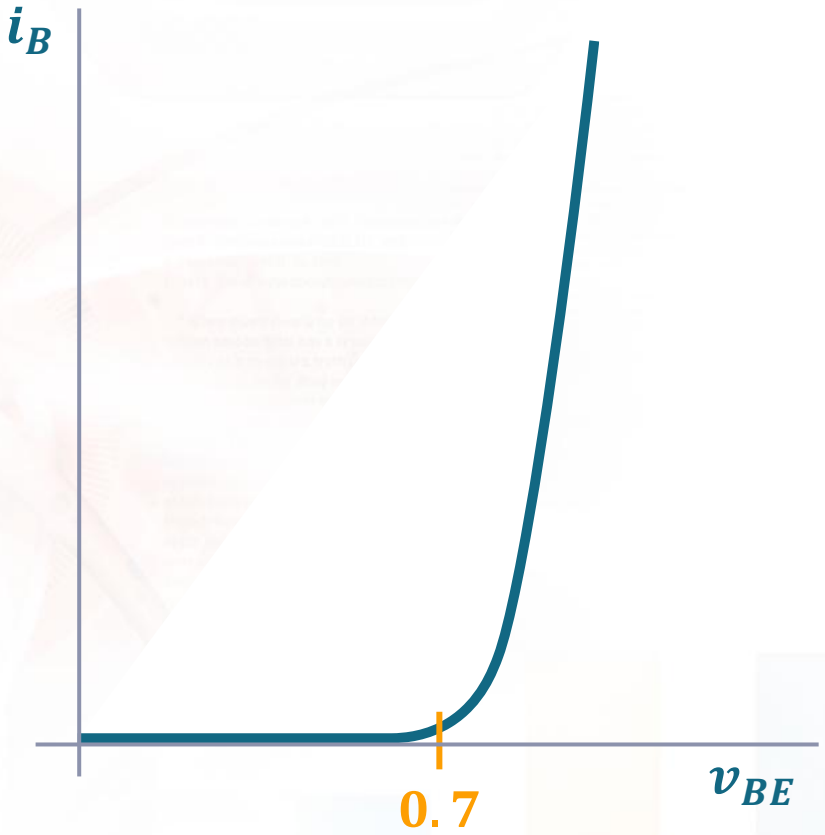




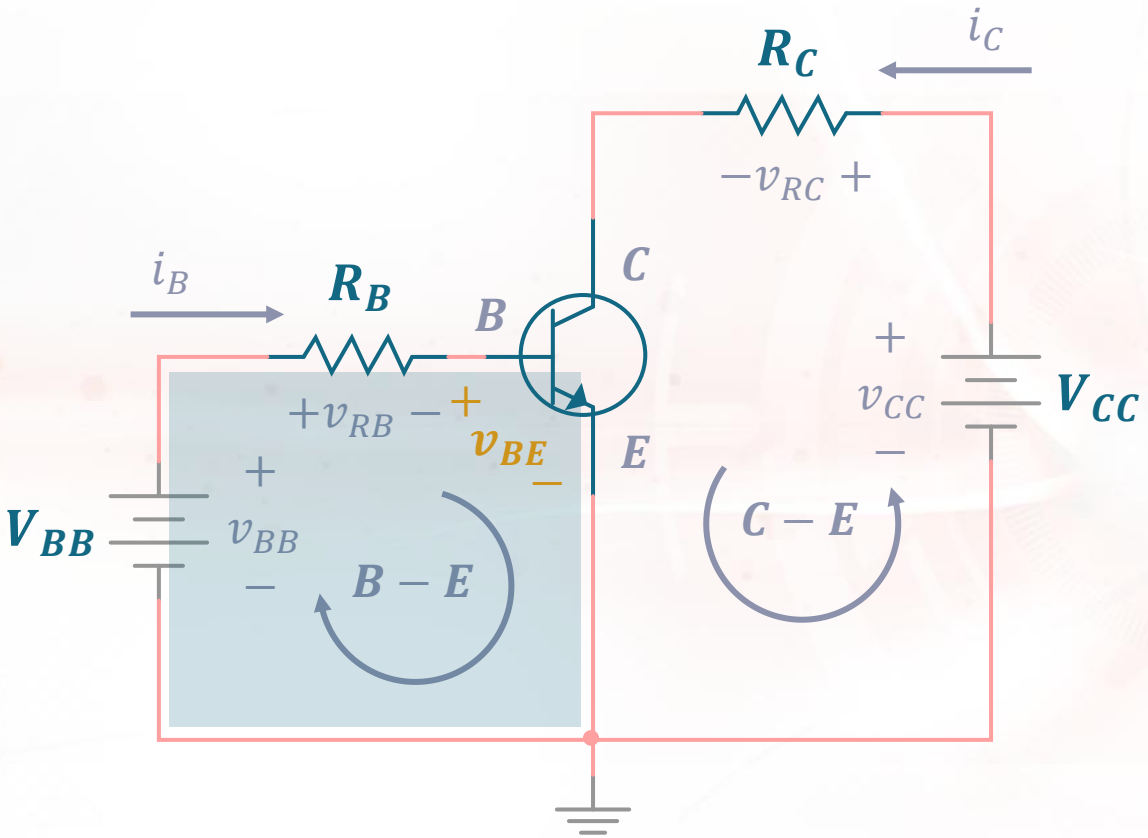
CHARACTERISTIC CURVE

THE BASE CURVE





BASE CURRENT



Base-Emitter Loop

$$-v_{BB} + v_{RB} + v_{BE} = 0$$

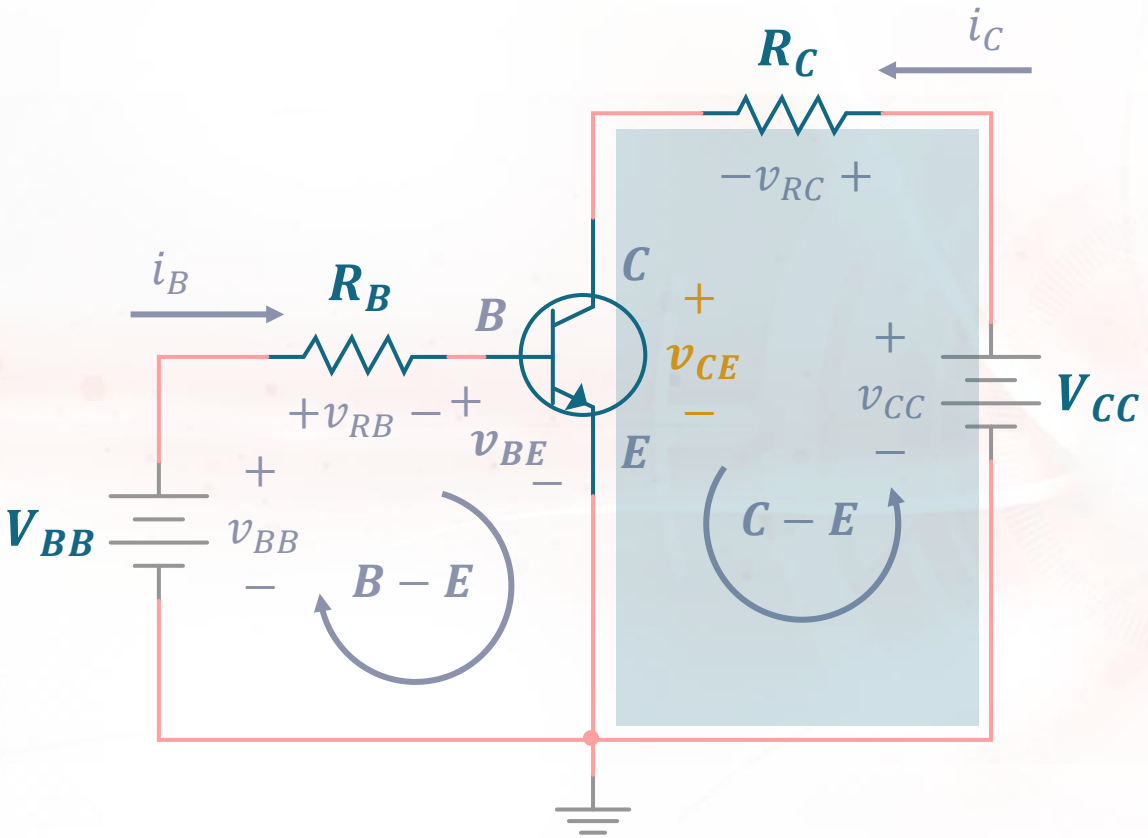
$$v_{RB} = v_{BB} - v_{BE}$$

$$i_B R_B = v_{BB} - v_{BE}$$

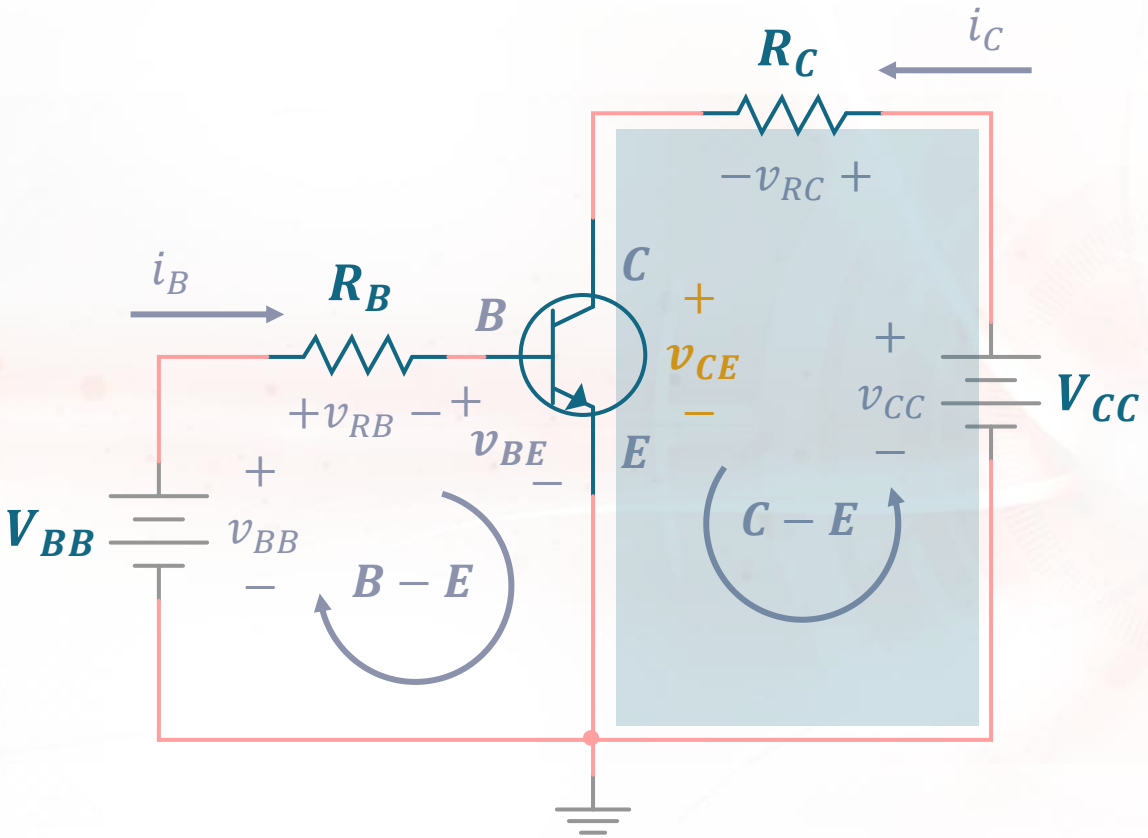
$$\boxed{i_B = \frac{V_{BB} - v_{BE}}{R_B}}$$

CE

THE COLLECTOR CURVE



COLLECTOR-EMITTER VOLTAGE



Collector-Emitter Loop

$$-v_{CE} - v_{RC} + v_{CC} = 0$$

$$v_{CE} = v_{CC} - v_{RC}$$

$$\boxed{v_{CE} = v_{CC} - i_C R_C} \quad \text{CE}$$

CURRENT GAIN

Alpha

$$\alpha_{dc} = \frac{i_C}{i_E}$$

$$\alpha_{dc} \approx 1$$

Beta

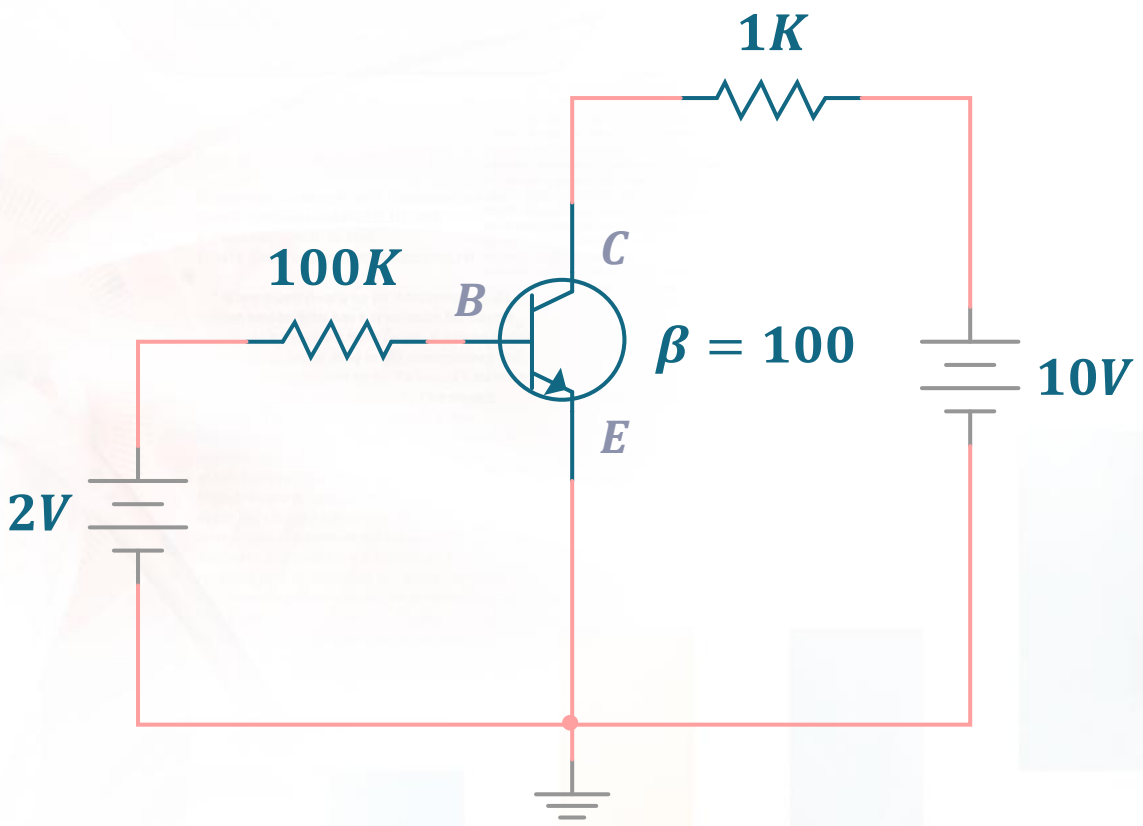
$$\beta_{dc} = \frac{i_C}{i_B}$$

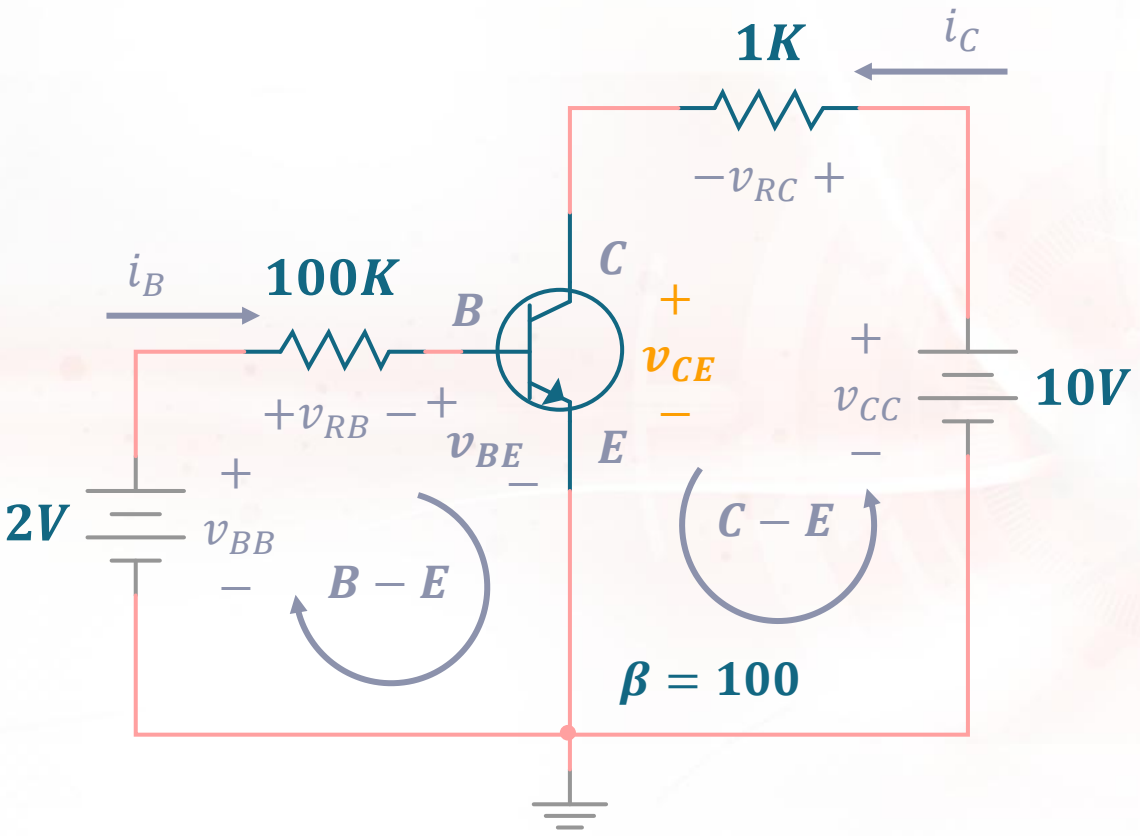
Determine the **parameters** of the given transistor network.

i_B, i_C, and v_{CE}

i_c @ $v_{CE} = 0$

v_{CE} @ $i_C = 0$





KVL @ $B - E$

$$-v_{BB} + v_{RB} + v_{BE} = 0$$

$$v_{RB} = v_{BB} - v_{BE}$$

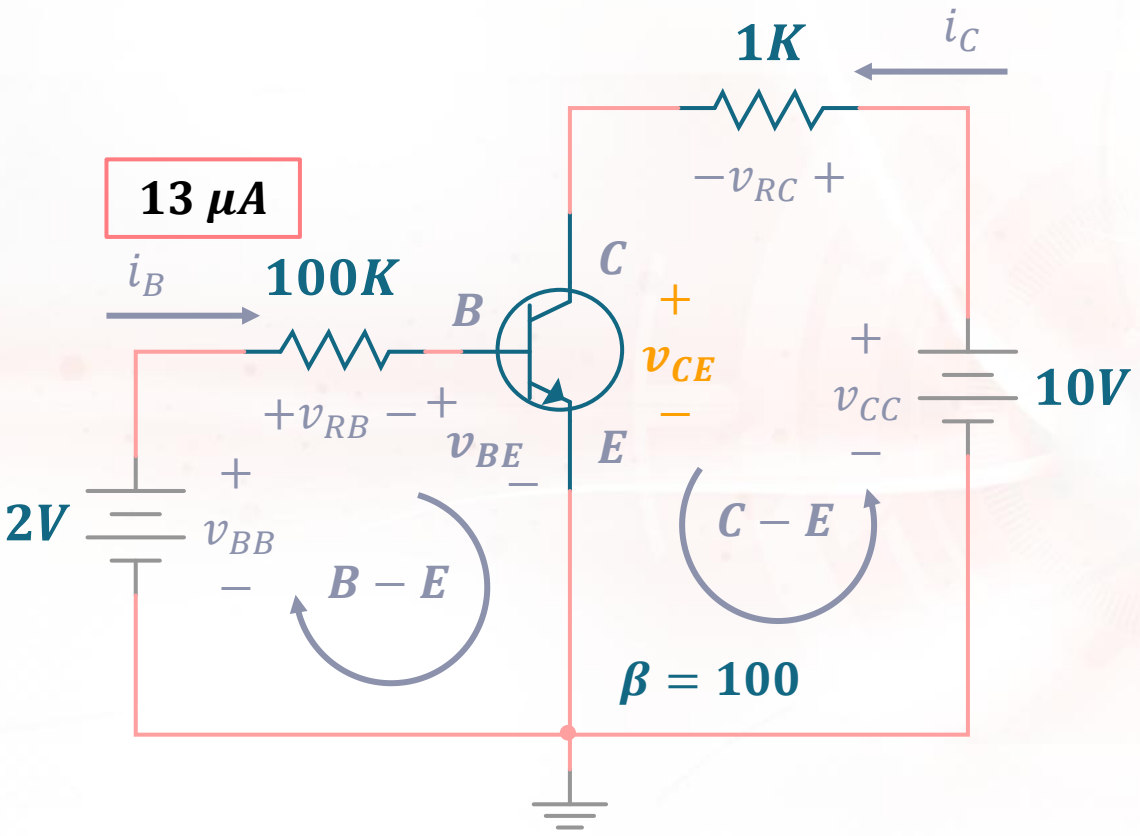
$$i_B R_B = v_{BB} - v_{BE}$$

$$i_B = \frac{v_{BB} - v_{BE}}{R_B}$$

$$i_B = \frac{2 - 0.7}{100K}$$

$$\boxed{i_B = 13\ \mu A}$$

ans



Current Gain

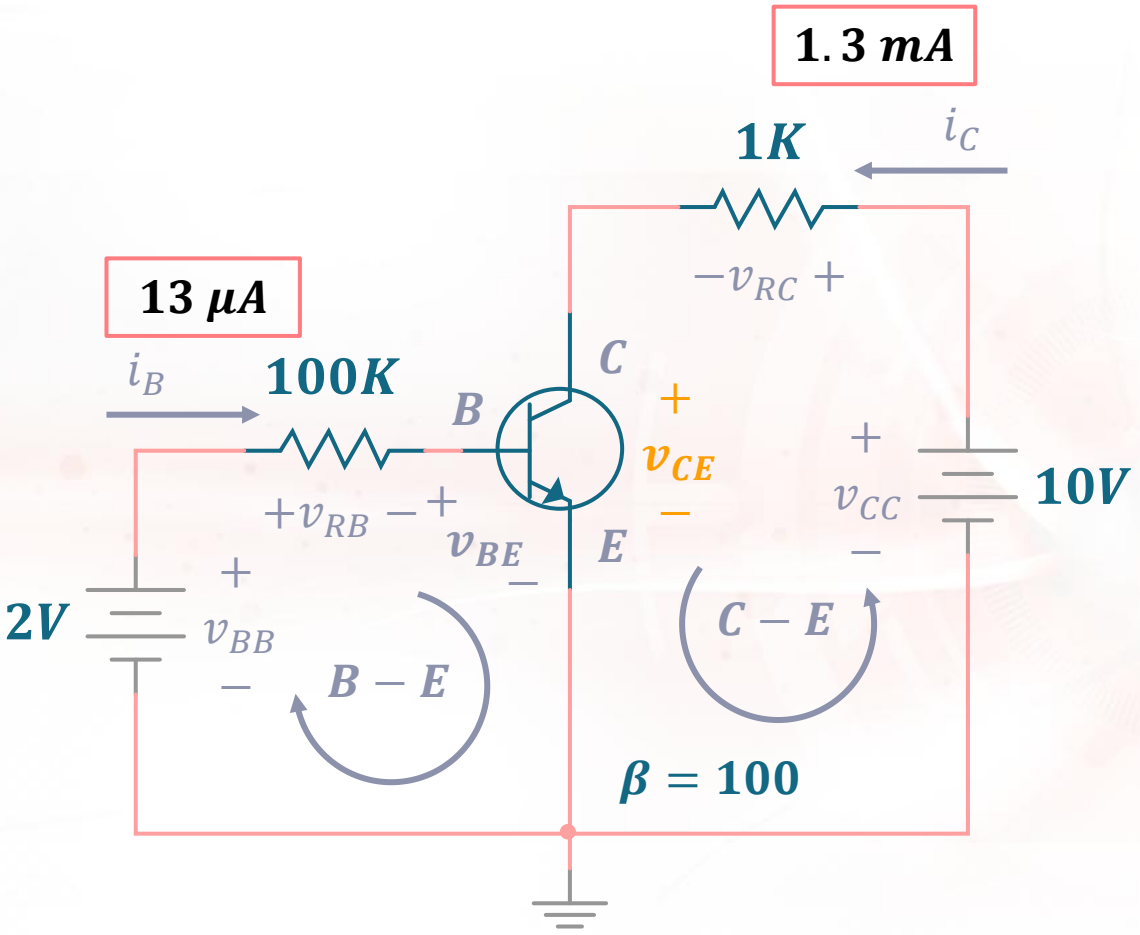
$$\beta = \frac{i_C}{i_B}$$

$$i_C = \beta i_B$$

$$i_C = 100(13\ uA)$$

$$\boxed{i_C = 1.3\ mA}$$

ans



KVL @ $C - E$

$$-v_{CE} - v_{RC} + v_{CC} = 0$$

$$v_{CE} = v_{CC} - v_{RC}$$

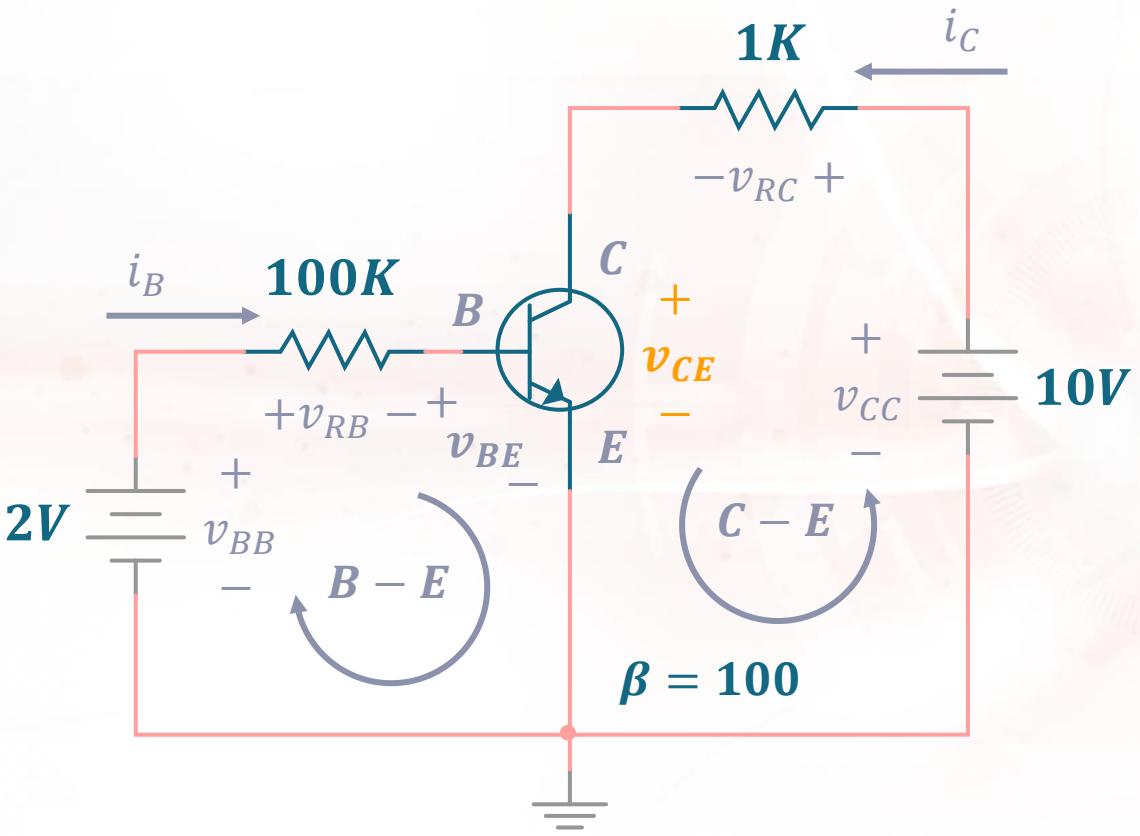
$$v_{CE} = v_{CC} - i_C R_C$$

$$v_{CE} = 10 - 1.3\ mA(1K)$$

$$\boxed{v_{CE} = 8.7\ V}$$

ans

v_{CE} @ $i_C = 0$



KVL @ $C - E$

$$-v_{CE} - v_{RC} + v_{CC} = 0$$

$$v_{CE} = v_{CC} - v_{RC}$$

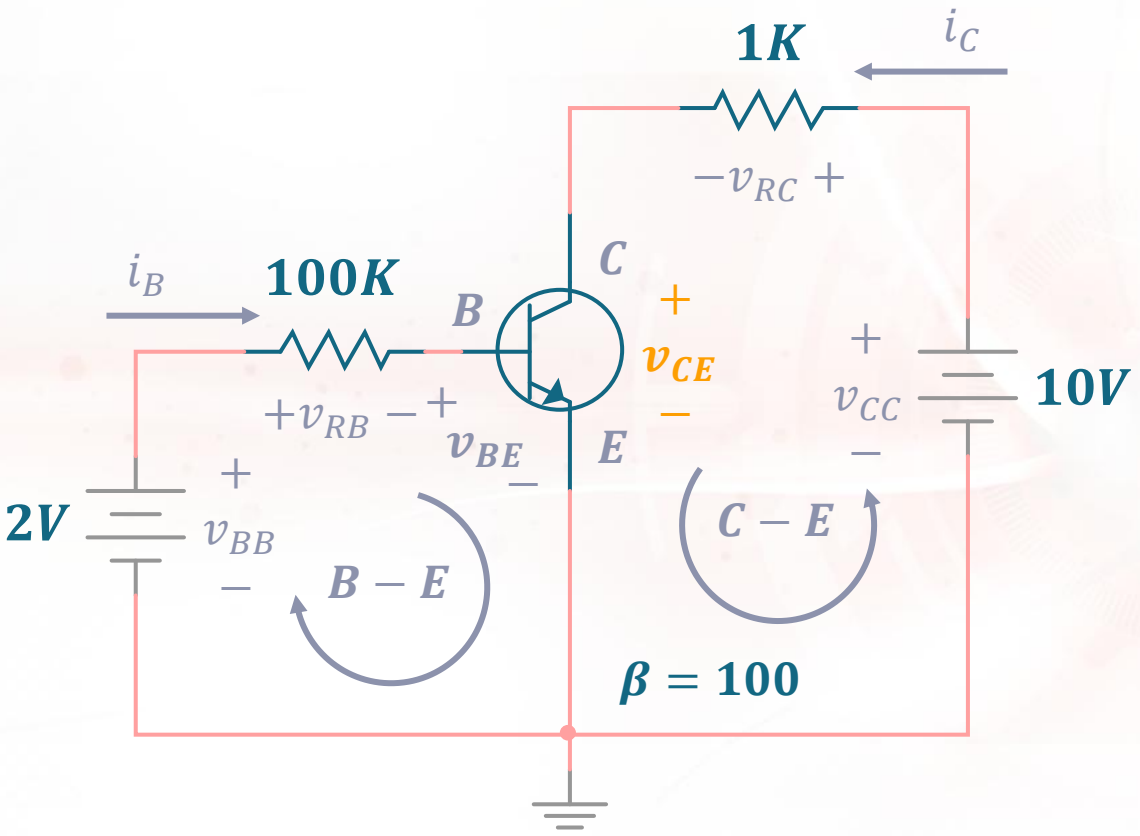
$$v_{CE} = v_{CC} - \cancel{i_C R_C}^{\,0}$$

$$v_{CE} = v_{CC}$$

$$\boxed{v_{CE(cutoff)} = 10\ V}$$

ans

i_c @ $v_{CE} = 0$



KVL @ $C - E$

$$-v_{CE} - v_{RC} + v_{CC} = 0$$

$$v_{CE} = v_{CC} - v_{RC} \quad (v_{CE} \to 0)$$

$$v_{RC} = v_{CC}$$

$$i_C R_C = v_{CC}$$

$$i_C = \frac{v_{CC}}{R_C}$$

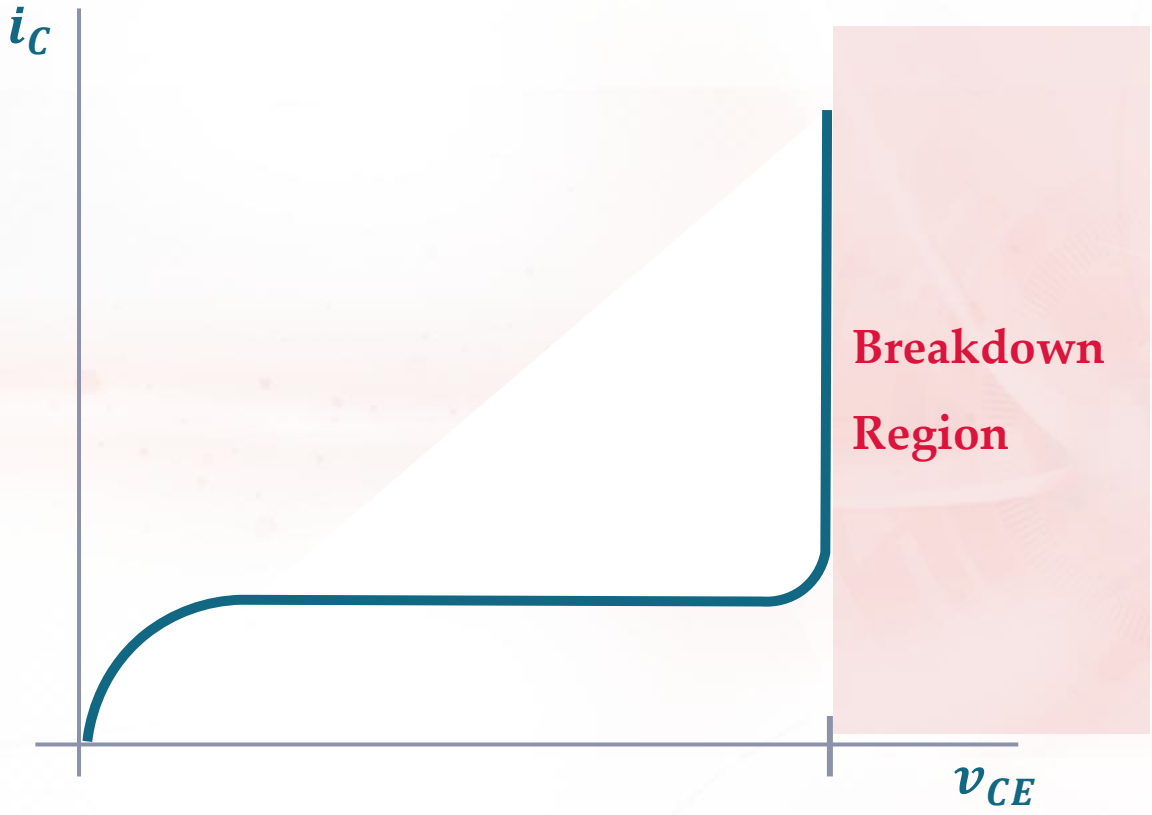
$$i_C = \frac{10}{1K}$$

$$\boxed{i_{C(sat)} = 10\ mA}$$

ans

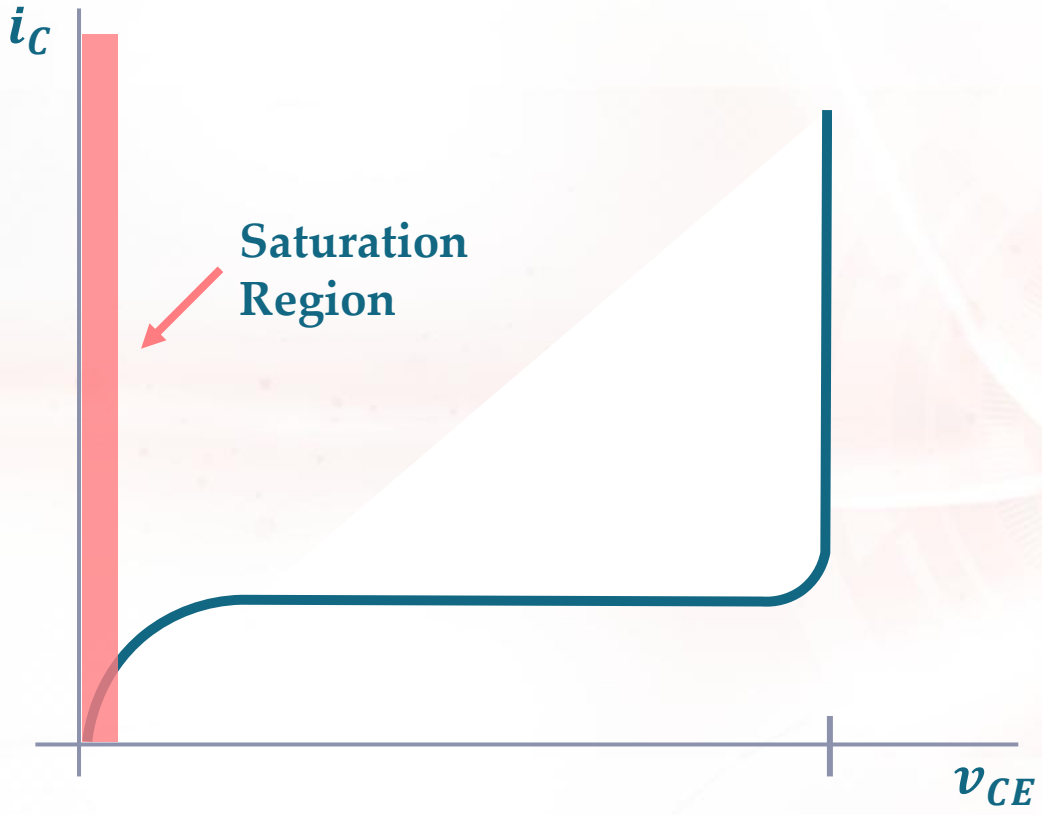
REGIONS OF OPERATION

BREAKDOWN REGION



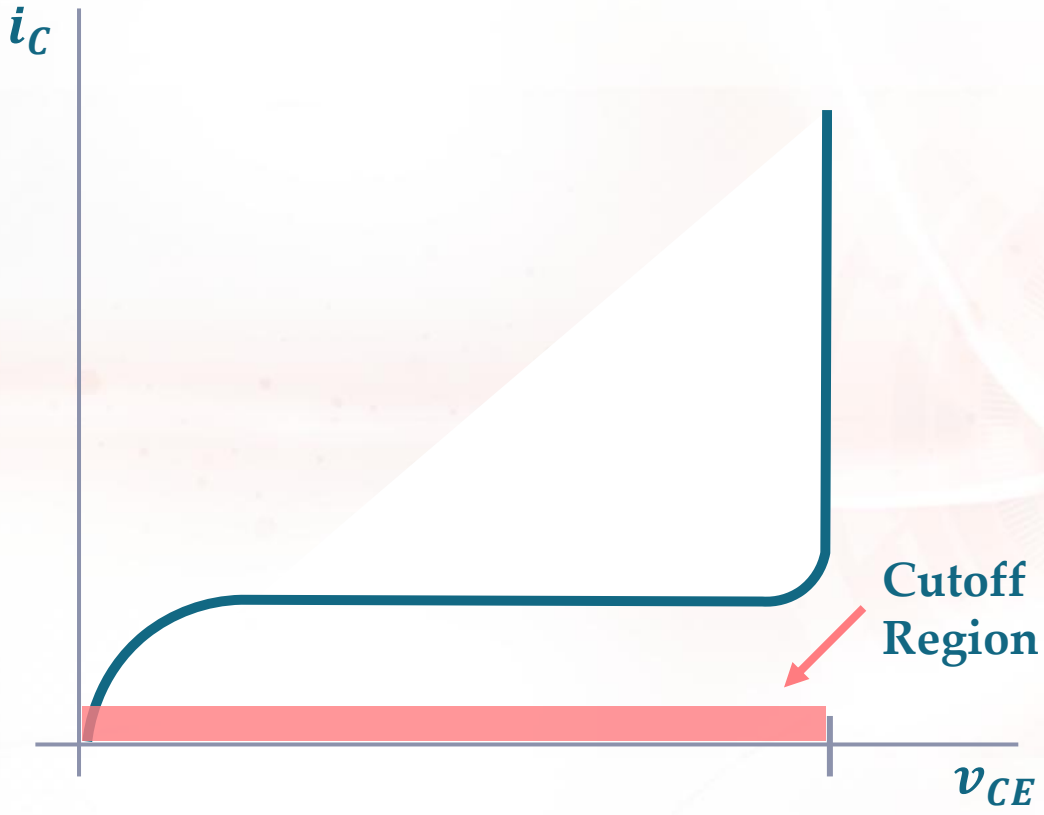
The transistor should never operate in this region because it will be **destroyed**

SATURATION REGION



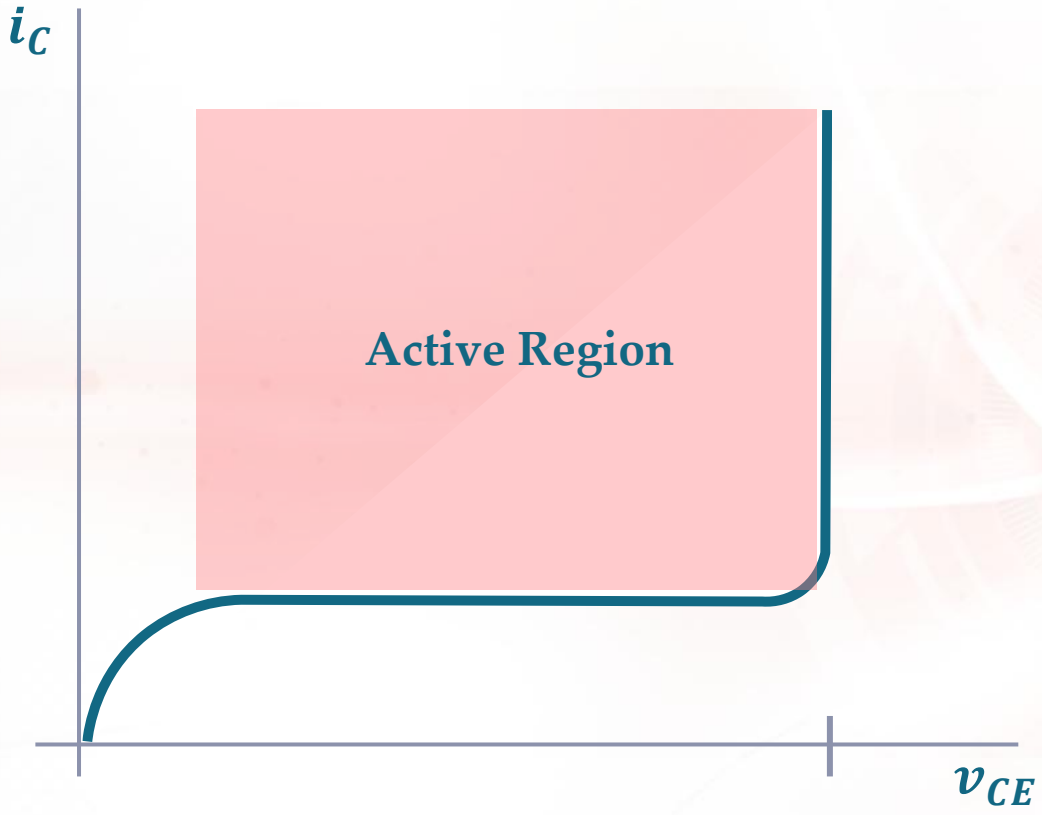
The early rising part of the curve, where v_{CE} is between **0V** and few tenths of a volt

CUTOFF REGION



A **small collector current** when the base current is zero

ACTIVE REGION

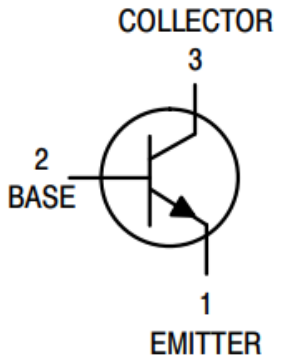


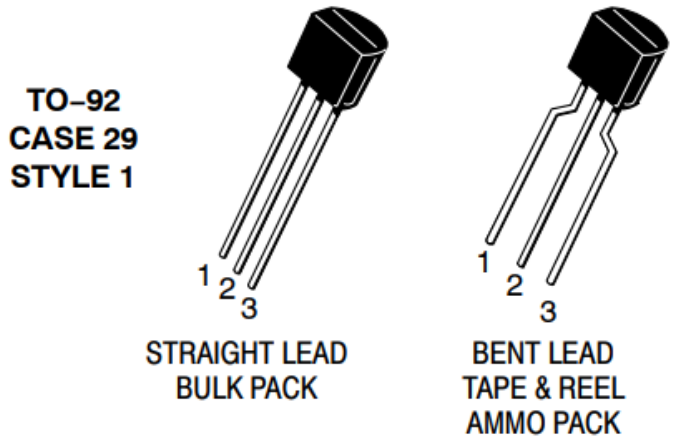
Collector current is **constant** in this region

READING DATASHEET

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Collector – Emitter Voltage	V_{CEO}	40	Vdc
Collector – Base Voltage	V_{CBO}	60	Vdc
Emitter – Base Voltage	V_{EBO}	6.0	Vdc
Collector Current – Continuous	I_C	200	mAdc
Total Device Dissipation @ T_A = 25°C Derate above 25°C	P_D	 625 5.0	 mW mW/°C
Total Device Dissipation @ T_C = 25°C Derate above 25°C	P_D	 1.5 12	 W mW/°C
Operating and Storage Junction Temperature Range	T_J, T_{stg}	–55 to +150	°C





ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)

Characteristic		Symbol	Min	Max	Unit
ON CHARACTERISTICS					
DC Current Gain (Note 2)		h_{FE}			–
(I_C = 0.1 mAdc, V_{CE} = 1.0 Vdc)	2N3903		20	–	
	2N3904		40	–	
(I_C = 1.0 mAdc, V_{CE} = 1.0 Vdc)	2N3903		35	–	
	2N3904		70	–	
(I_C = 10 mAdc, V_{CE} = 1.0 Vdc)	2N3903		50	150	
	2N3904		100	300	
(I_C = 50 mAdc, V_{CE} = 1.0 Vdc)	2N3903		30	–	
	2N3904		60	–	
(I_C = 100 mAdc, V_{CE} = 1.0 Vdc)	2N3903		15	–	
	2N3904		30	–	
Collector – Emitter Saturation Voltage (Note 2)		$V_{CE(sat)}$			Vdc
(I_C = 10 mAdc, I_B = 1.0 mAdc)			–	0.2	
(I_C = 50 mAdc, I_B = 5.0 mAdc			–	0.3	
Base – Emitter Saturation Voltage (Note 2)		$V_{BE(sat)}$			Vdc
(I_C = 10 mAdc, I_B = 1.0 mAdc)			0.65	0.85	
(I_C = 50 mAdc, I_B = 5.0 mAdc)			–	0.95	

LABORATORY